

Physics-Informed Reconstruction of Cylinder Flow with Reynolds Continuation

PINN-based reconstruction and transition analysis for 2D incompressible flow past a cylinder

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Project Overview

Physical problem, objectives, and expected transition behavior

Problem Statement

- 2D incompressible flow past a circular cylinder is a canonical benchmark in fluid dynamics.
- As Reynolds number increases, the wake evolves from steady symmetric recirculation to time-periodic vortex shedding.
- This project uses a Physics-Informed Neural Network (PINN) to reconstruct the flow field and study the transition region.

Geometry and setup

Cylinder radius $R = 0.5$

Domain x in $[-5, 25]$, y in $[-10, 10]$

Target Reynolds sweep: $Re = 5$ to 100

Main Deliverables

Flow reconstruction

Velocity, pressure, vorticity, and streamlines

Reusable pipeline

Continuation training across Reynolds numbers

Governing Equations and PINN Formulation

2D incompressible Navier-Stokes with stream-function parameterization

Navier-Stokes Equations

$$\begin{aligned}u_t + u u_x + v u_y &= -p_x + (1/\text{Re}) \\&\quad (u_{xx} + u_{yy}) \\v_t + u v_x + v v_y &= -p_y + (1/\text{Re}) \\&\quad (v_{xx} + v_{yy}) \\u_x + v_y &= 0\end{aligned}$$

The PINN minimizes PDE residuals together with boundary, initial, and outlet constraints using automatic differentiation.

Stream Function Form

Represent the solution as
 $\psi(x, y, t), p(x, y, t)$

$$\begin{aligned}u &= d\psi / dy \\v &= -d\psi / dx\end{aligned}$$

- Enforces incompressibility by construction.
- Removes the need for an extra divergence penalty term.
- Produces smooth derivatives compatible with PDE residual evaluation.

Network Architecture, Training, and Loss Design

Model Architecture

$(x, y, t) \rightarrow \text{MLP}$
 $\rightarrow (\psi, p)$

Width 128 neurons/layer

Depth 6 hidden layers

Activation
n tanh

Optimizer
r Adam + L-BFGS

Loss Function

$$L = L_{\text{pde}} + 10 L_{\text{bc}} + 10 L_{\text{ic}} + L_{\text{out}}$$

- PDE loss: momentum residuals from Navier-Stokes.
- Boundary loss: inlet, wall, and far-field conditions.
- Initial loss: uniform initialization in the domain.
- Outlet loss: pressure / gradient regularization downstream.

Sampling Strategy

Physics-aware collocation

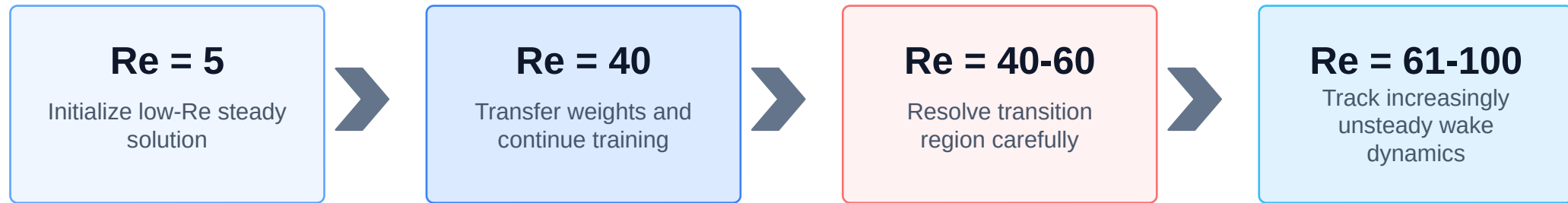


- Refine points near the cylinder to capture boundary-layer behavior.
- Bias sampling in the wake to resolve separation and shedding.
- Use smoother activation and derivatives for stable AD-based training.

Reynolds Continuation Strategy

Warm-starting across Reynolds numbers to improve optimization stability

Continuation Workflow



More stable optimization

Fewer iterations at each new
Re

Better continuity across the
parameter sweep

Results Placeholder: Flow Reconstruction

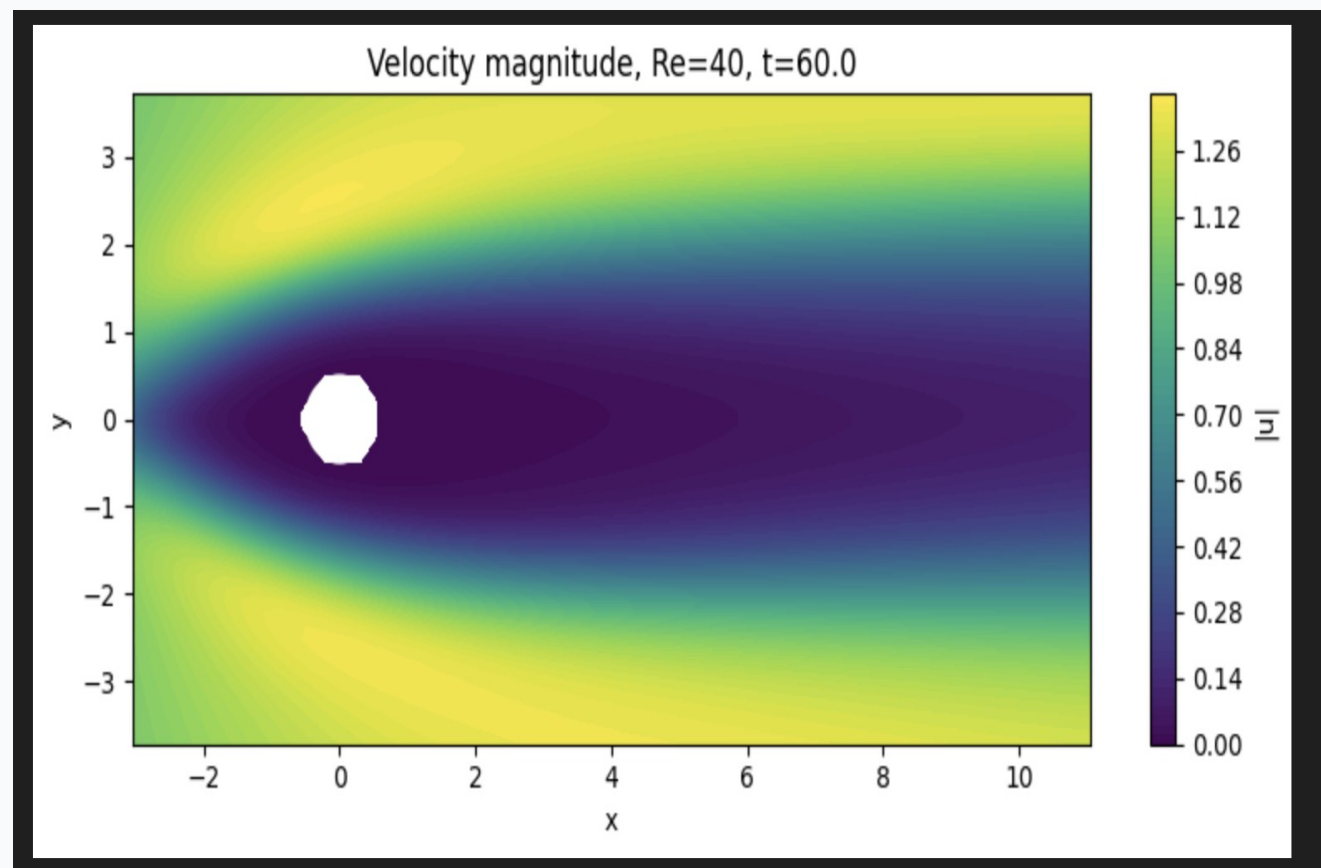
Replace placeholder panels with your exported figures

Key observations

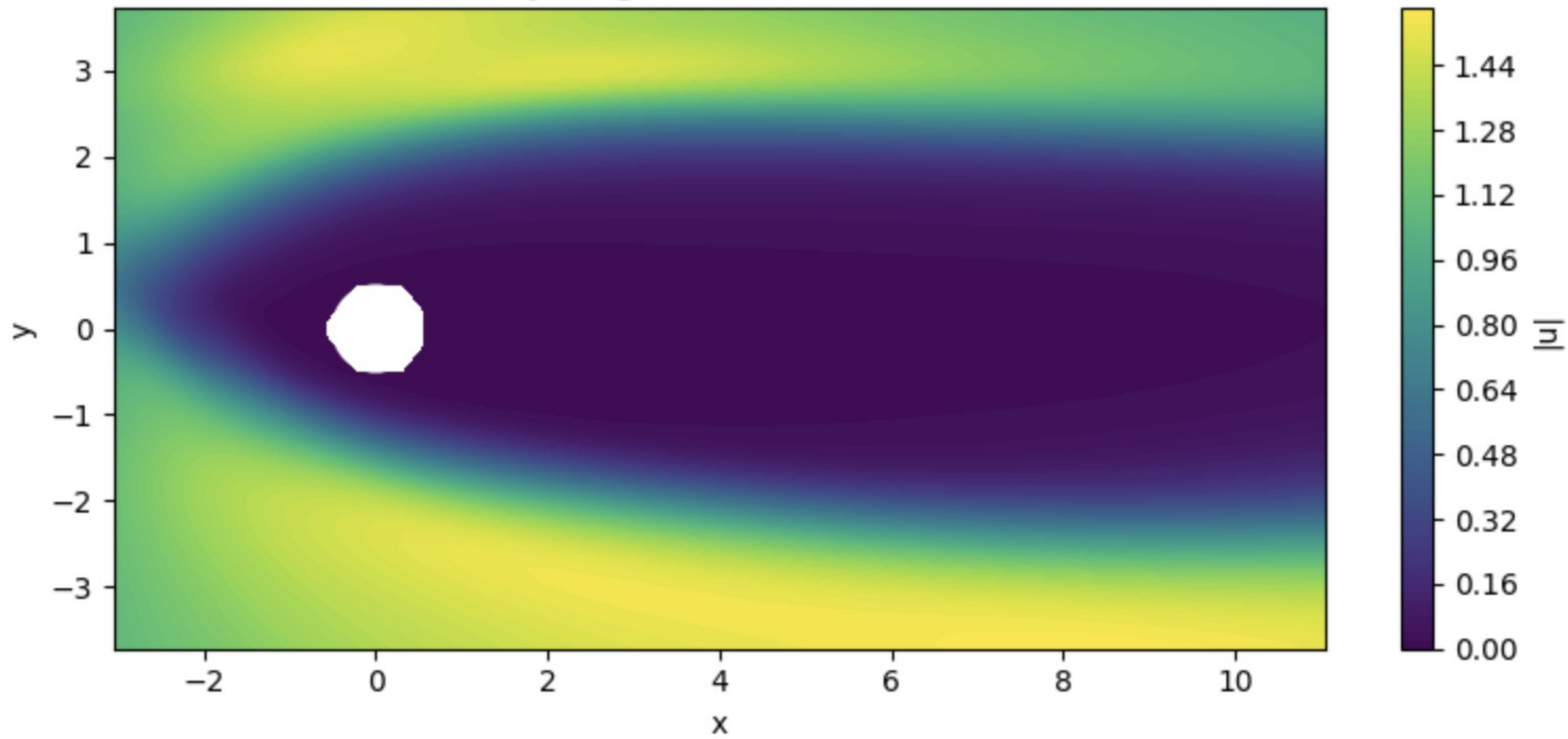
- Velocity magnitude should show stagnation at the front face and acceleration near the cylinder sides.
- Downstream recovery and recirculation bubble length can be compared across Reynolds numbers.
- Use the same colorbar range when comparing multiple Re values.

Recommended figures

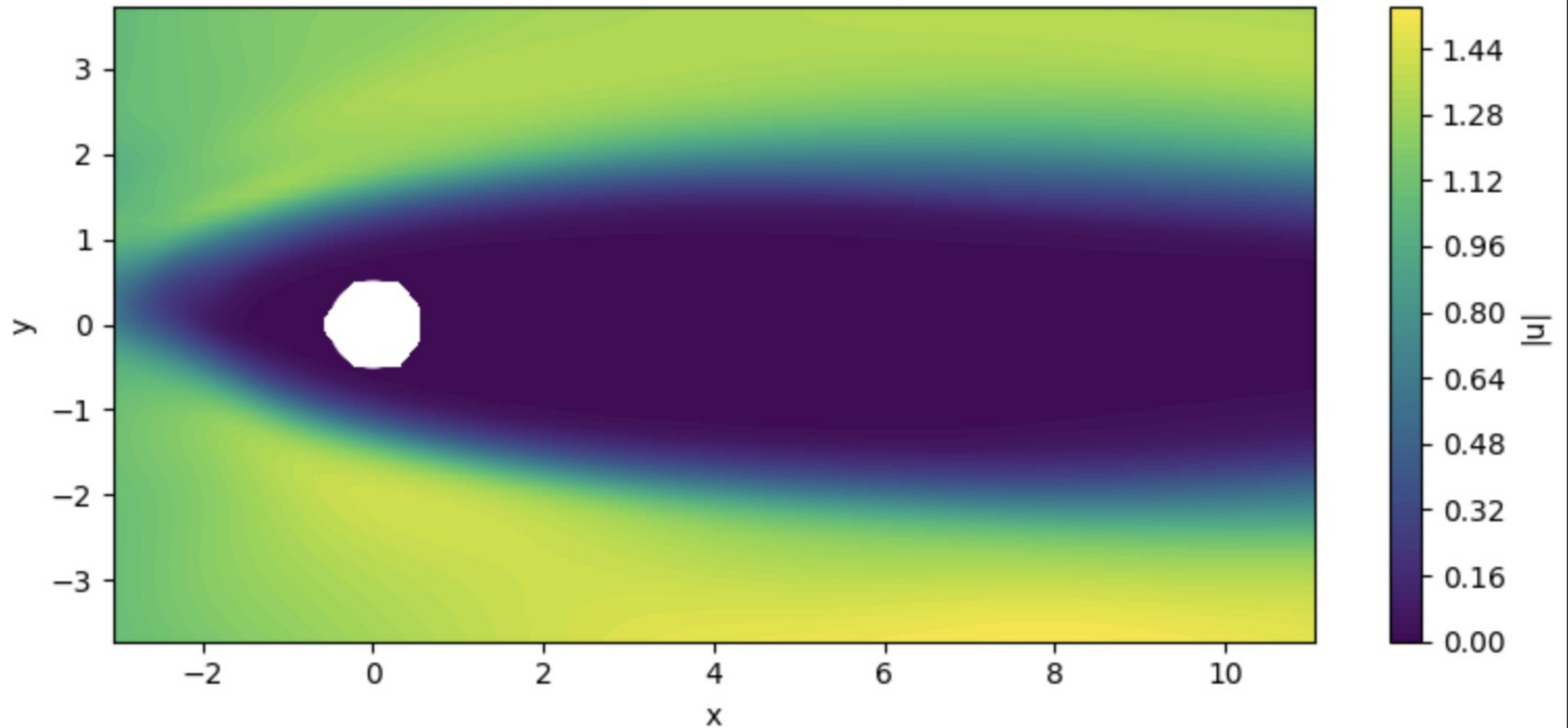
- Velocity magnitude contour at representative times
- Streamline plot showing wake topology
- Optional comparison: $Re = 40, 60, 100$



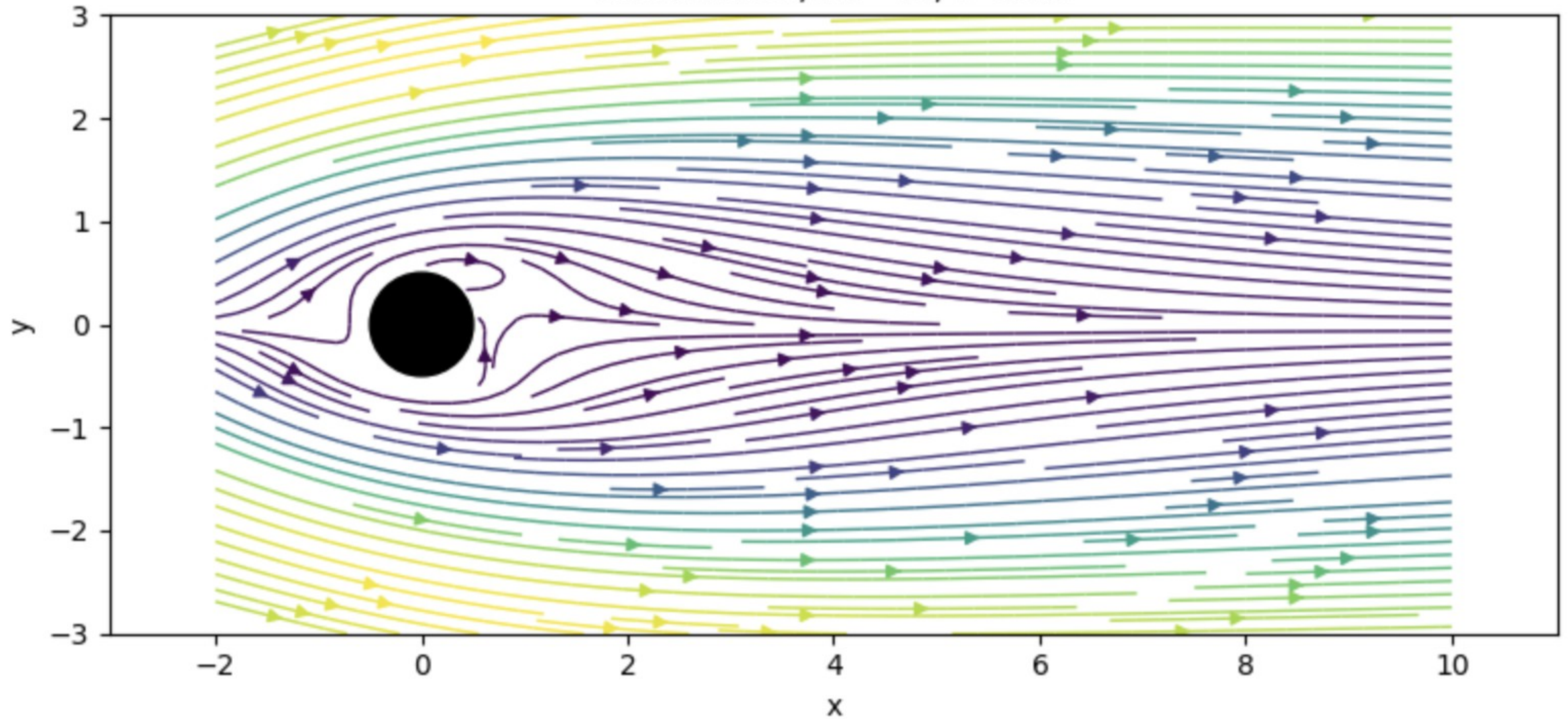
Velocity magnitude, $Re=60$, $t=60.0$



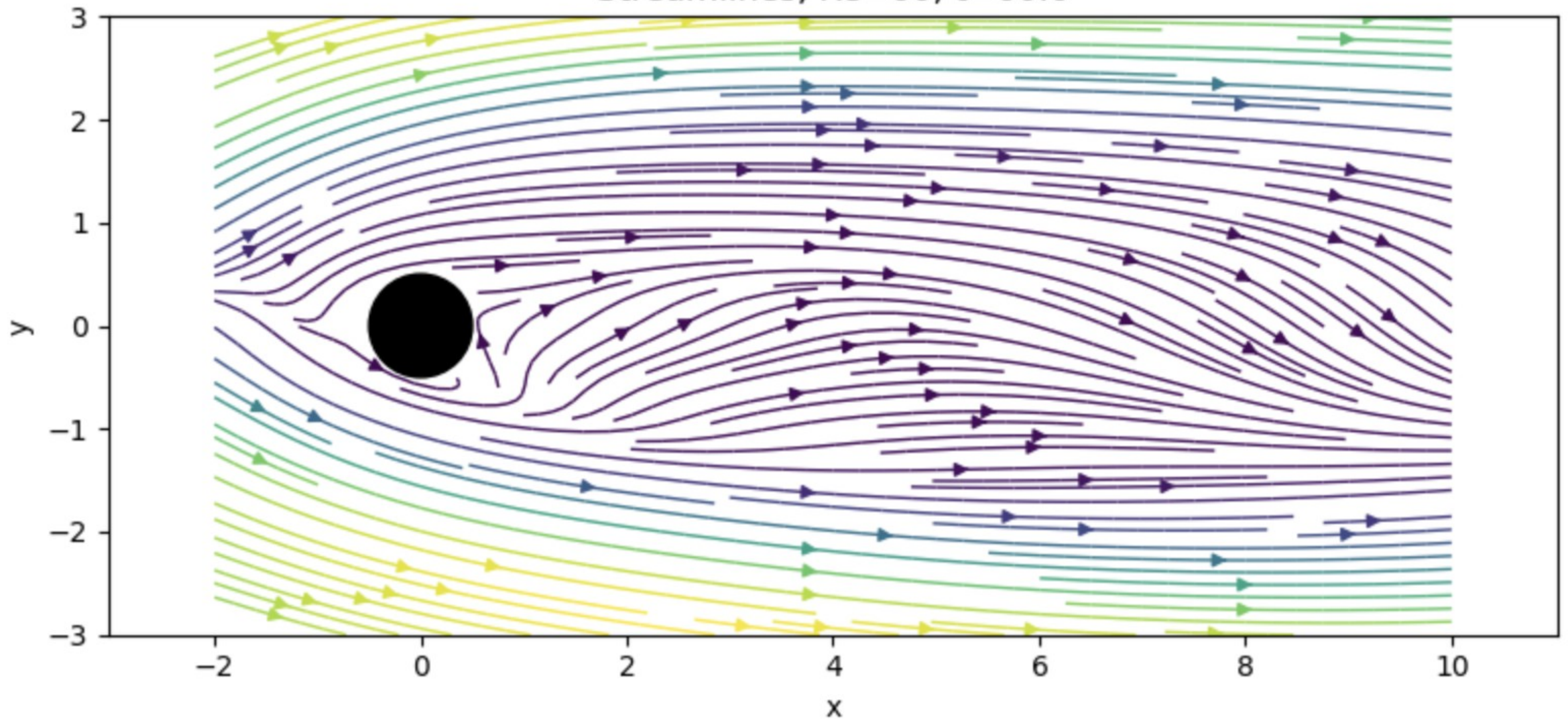
Velocity magnitude, $Re=100$, $t=60.0$



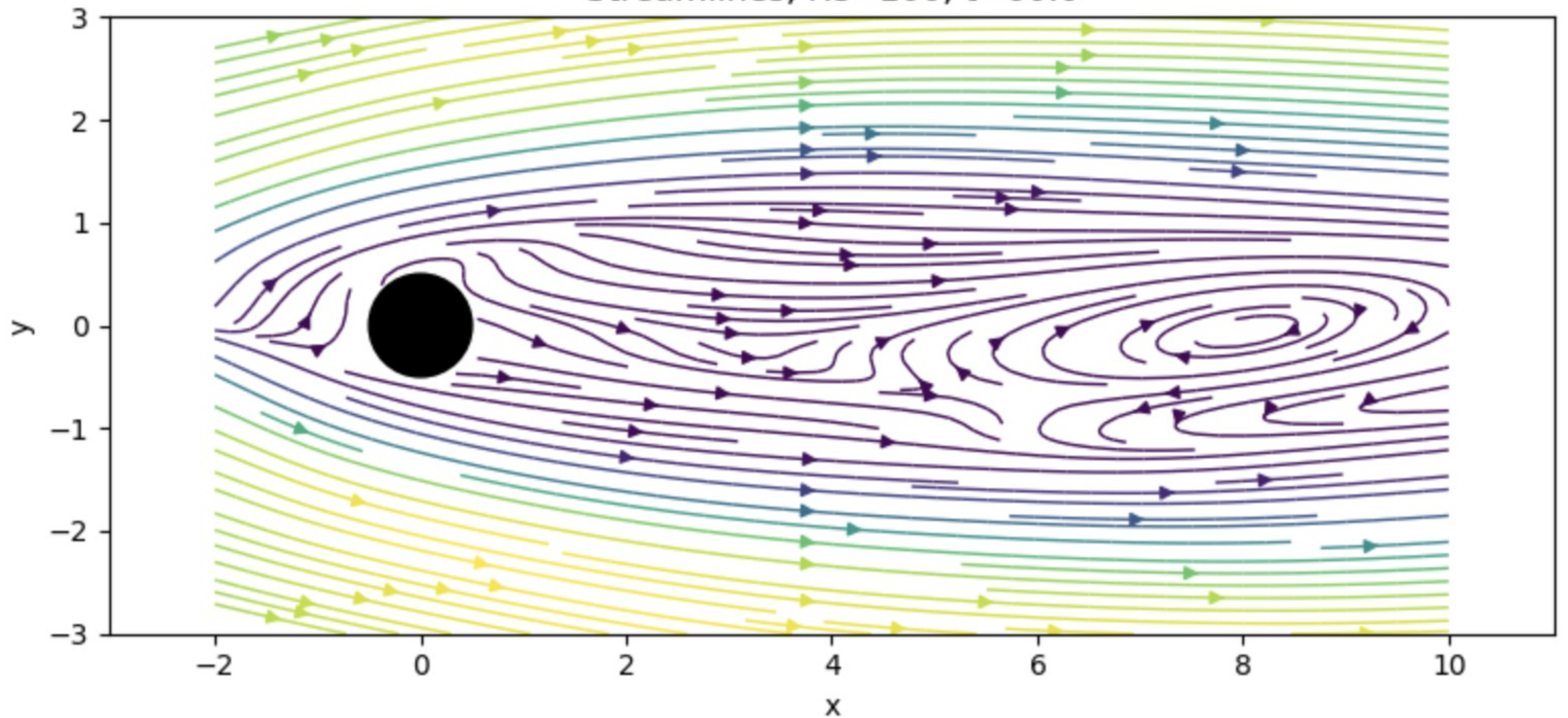
Streamlines, $Re=40$, $t=60.0$



Streamlines, $Re=60$, $t=60.0$

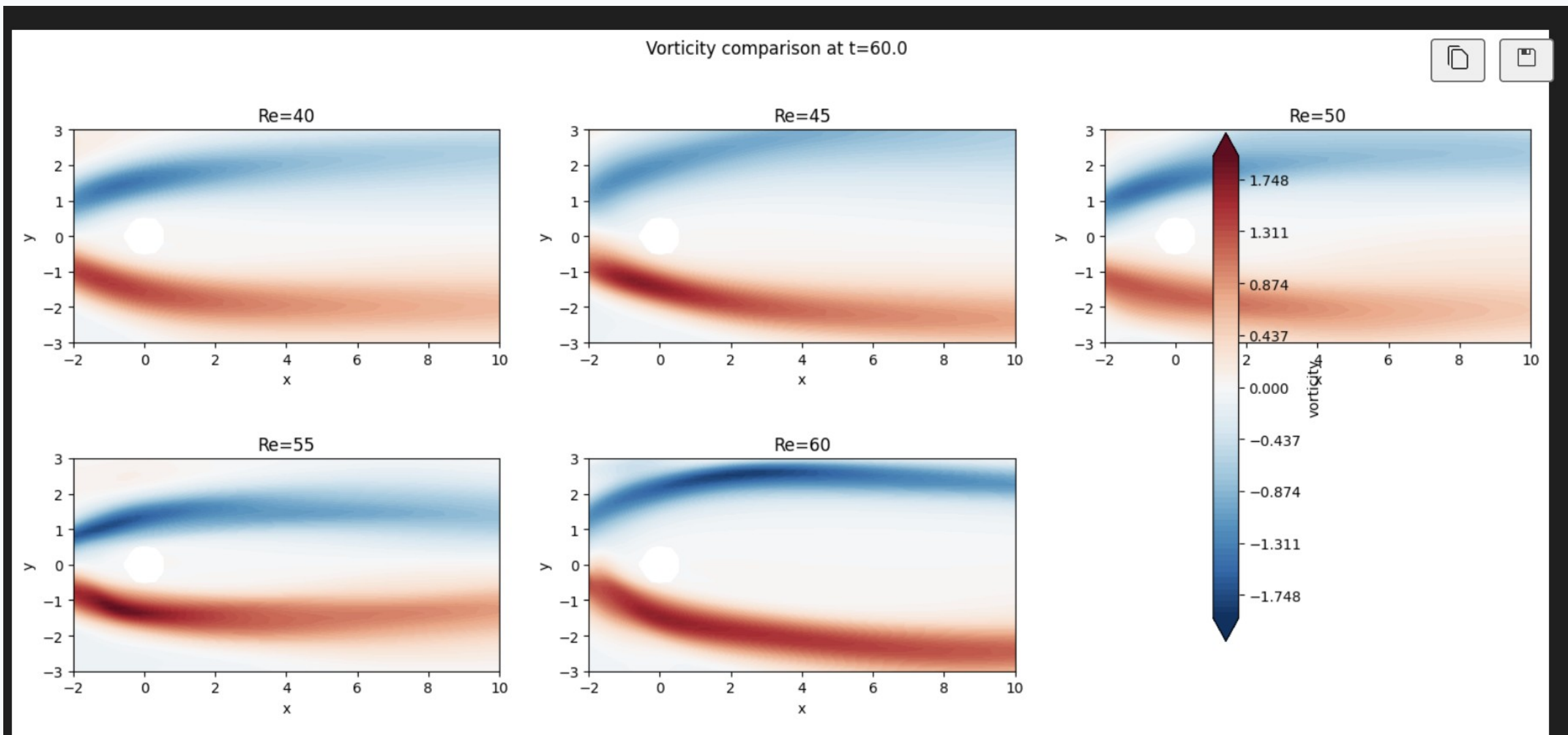


Streamlines, $Re=100$, $t=60.0$

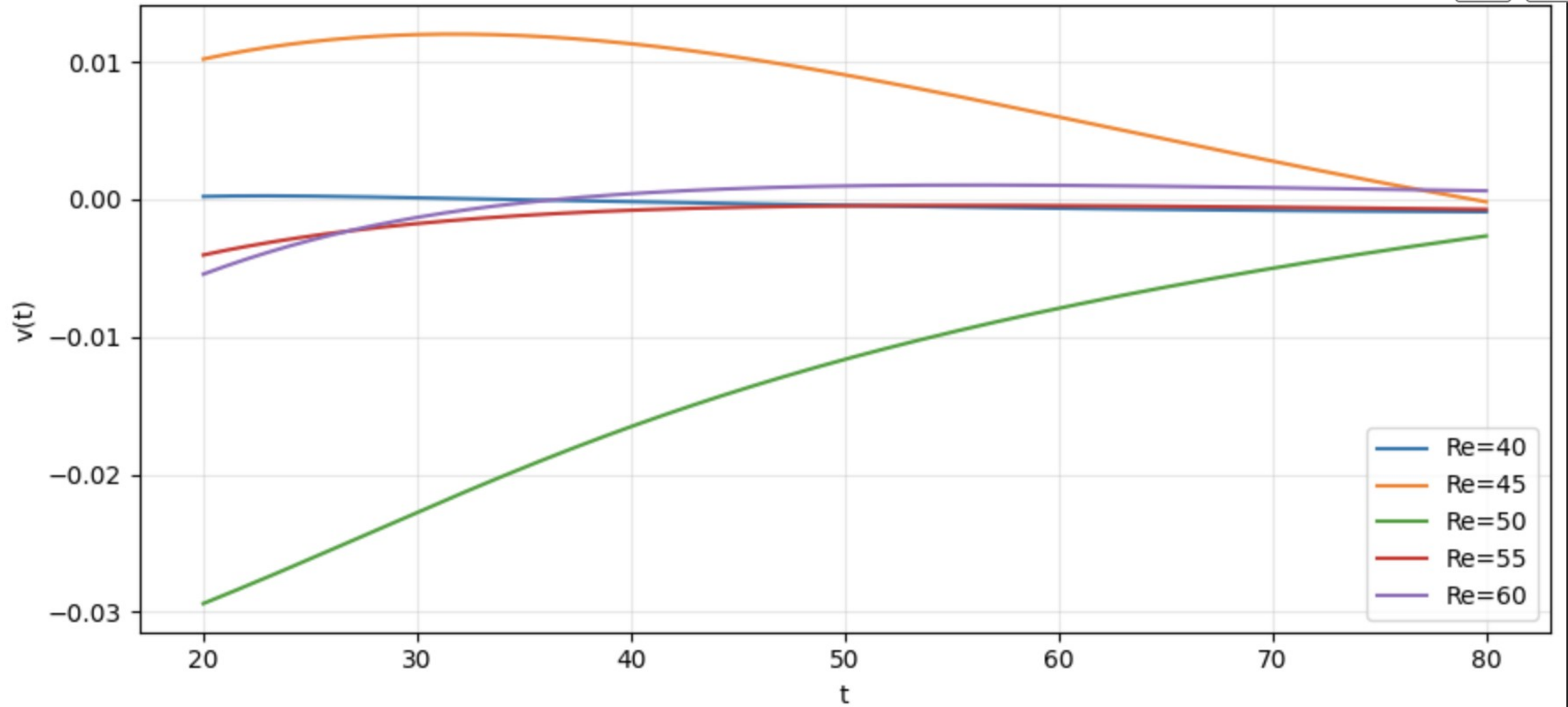


Results Placeholder: Vorticity and Probe Signals

Use these for wake structure and time-series evidence of transition

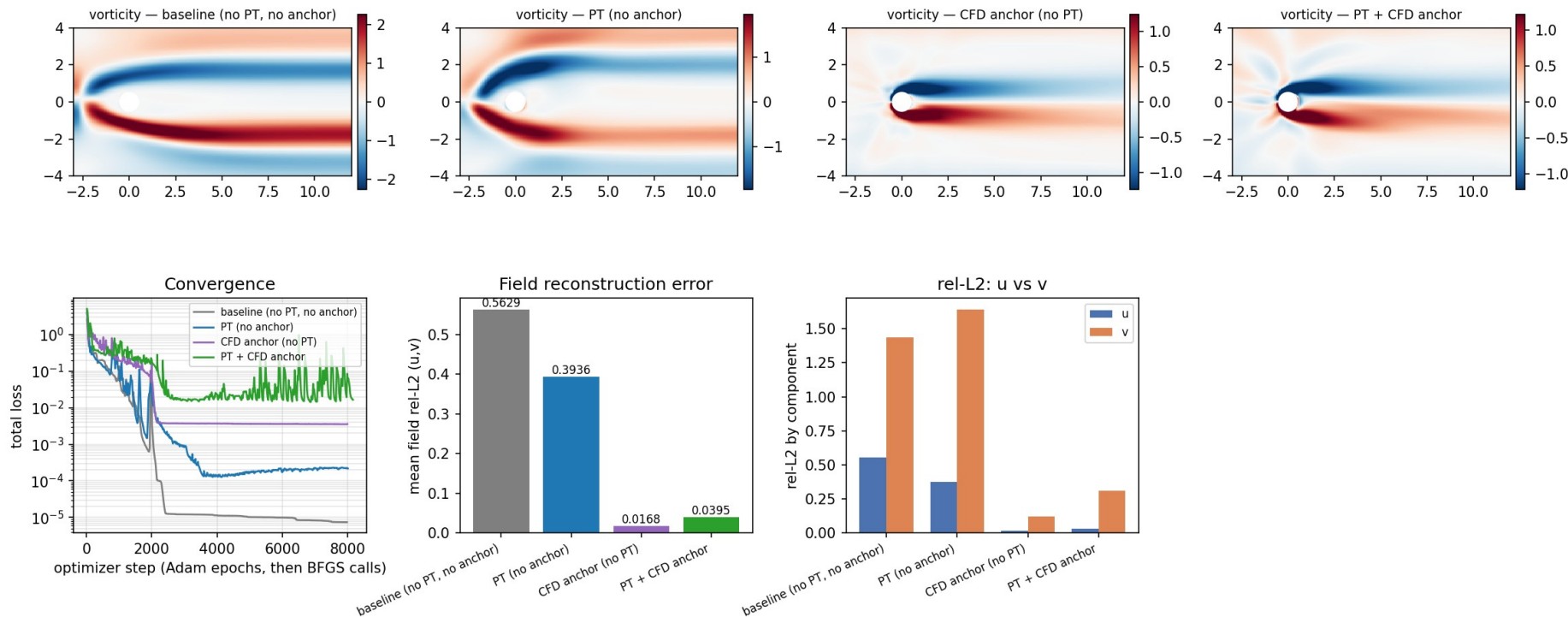


Probe signal at (3.0, 0.0), component=v



Ablation: pseudo-time (PT) x CFD data anchor (Re=40, steady)

Ablation (Re=40, steady): full recipe vs removing each ingredient



variant	rel-L2 (u,v)	rel-L2 u	rel-L2 v
baseline (no PT, no anchor)	0.5629	0.5549	1.4357
PT (no anchor)	0.3936	0.3767	1.6424
CFD anchor (no PT)	0.0168	0.0145	0.1195
PT + CFD anchor	0.0395	0.0327	0.3117

Takeaway: the CFD data anchor is the dominant lever (field rel-L2 0.017). Physics-only PINNs (no anchor) reach the lowest training loss yet the WORST field — a spurious wake

What the 2x2 shows, and how to reach ground truth

Key findings

CFD data anchor is decisive: field rel-L2 (u,v) 0.56 -> 0.017. Best cell = anchor, NO PT.

Physics-only PINN (no anchor) finds a SPURIOUS wake: lowest training loss ($7e-6$) but worst field (0.56).

PT helps only without data: 0.56 -> 0.39 (avoids the spurious solution, per Wang et al. 2026) -- but still far from truth.

PT + anchor was WORSE than anchor alone (0.040 vs 0.017): PT weight saturated at the ceiling and fought the data.

Next steps -> full field ~ ground truth

1. Keep the full-field CFD anchor -- it is what gets you to ~1.7%. This is the working config.
2. Fix the wake / v-component (rel-L2 v still 0.12): add Fourier features + hard BCs (cfp40_v2), and more collocation + data in the near wake.
3. Anchor pressure (pMean) too, not just u/v, for the full physical field.
4. Drop PT in the data-supervised setting; reserve PT for data-free / forward solves. If kept, widen w_max and let the gamma shrink activate.
5. Tighter convergence: longer BFGS or the two-stage recipe (data-only pretrain -> PDE on).